P2 Chapter 8: Parametric Equations

Chapter Practice

Key Points

Summary of key points

1. A curve can be defined using parametric equations $x = p(t)$ and $y = q(t)$. Each value of the parameter, t, defines a point on the curve with coordinates $(p(t), q(t))$.

2. You can convert between parametric equations and Cartesian equations by using substitution to eliminate the parameter.

3. For parametric equations $x = p(t)$ and $y = q(t)$ with Cartesian equation $y = f(x)$:
   - the domain of $f(x)$ is the range of $p(t)$
   - the range of $f(x)$ is the range of $q(t)$

4. You can use parametric equations to model real-life situations. In mechanics you will use parametric equations with time as a parameter to model motion in two dimensions.

Chapter Exercises

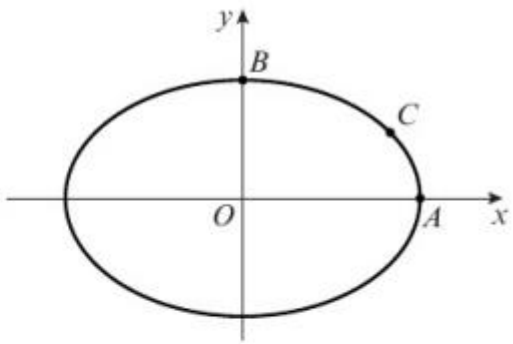
1 The diagram shows a sketch of the curve with parametric equations

$$x = 4\cos t, \quad y = 3\sin t, \quad 0 \leqslant t < 2\pi$$

a Find the coordinates of the points A and B.

b The point C has parameter $t = \frac{\pi}{6}$. Find the exact coordinates of C.

c Find the Cartesian equation of the curve.

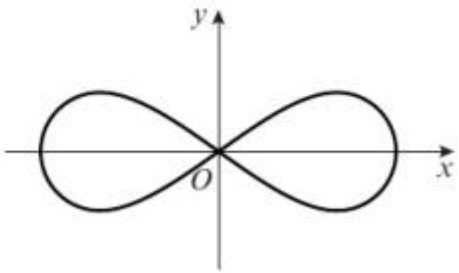


2 The diagram shows a sketch of the curve with parametric equations

$$x = \cos t, \quad y = \tfrac{1}{2}\sin 2t, \quad 0 \leqslant t < 2\pi$$

The curve is symmetrical about both axes.

Copy the diagram and label the points having parameters $t = 0$, $t = \frac{\pi}{2}$, $t = \pi$ and $t = \frac{3\pi}{2}$



Chapter Exercises

3 A curve has parametric equations

$$x = e^{2t+1} + 1, \quad y = t + \ln 2, \quad t > 1$$

a Find a Cartesian equation of this curve in the form $y = f(x)$, $x > k$ where k is a constant to be found in exact form.

b Write down the range of $f(x)$, leaving your answer in exact form.

4 A curve has parametric equations

$$x = \frac{1}{2t+1}, \quad y = 2\ln\left(t + \frac{1}{2}\right), \quad t > \frac{1}{2}$$

Find a Cartesian equation of the curve in the form $y = f(x)$, and state the domain and range of $f(x)$.

5 A curve has parametric equations $x = \sin t, \quad y = \cos 2t, \quad 0 \leqslant t < 2\pi$

a Find a Cartesian equation of the curve.

The curve cuts the x-axis at $(a, 0)$ and $(b, 0)$.

b Find the values of a and b.

6 A curve has parametric equations $x = \dfrac{1}{1+t}, \quad y = \dfrac{1}{(1+t)(1-t)}, \quad t > 1$

Express t in terms of x, and hence show that a Cartesian equation of the curve is $y = \dfrac{x^2}{2x-1}$

Chapter Exercises

7 A circle has parametric equations $x = 4\sin t - 3, \quad y = 4\cos t + 5, \quad 0 \leqslant t \leqslant 2\pi$

a Find a Cartesian equation of the circle.

b Draw a sketch of the circle.

c Find the exact coordinates of the points of intersection of the circle with the y-axis.

8 The curve C has parametric equations

$$x = \frac{2 - 3t}{1 + t}, \quad y = \frac{3 + 2t}{1 + t}, \quad 0 \leqslant t \leqslant 4$$

a Show that the curve C is part of a straight line. **(3 marks)**

b Find the length of this line segment. **(2 marks)**

9 A curve C has parametric equations

$$x = t^2 - 2, \quad y = 2t, \quad 0 \leqslant t \leqslant 2$$

a Find the Cartesian equation of C in the form $y = \mathrm{f}(x)$. **(3 marks)**

b State the domain and range of $y = \mathrm{f}(x)$ in the given domain of t. **(3 marks)**

c Sketch the curve in the given domain of t. **(2 marks)**

10 A curve C has parametric equations

$$x = 2\cos t, \quad y = 2\sin t - 5, \quad 0 \leqslant t \leqslant \pi$$

a Show that the curve C forms part of a circle. **(3 marks)**

b Sketch the curve in the given domain of t. **(3 marks)**

c Find the length of the curve in the given domain of t. **(3 marks)**

Chapter Exercises

11 The curve C has parametric equations

$$x = t - 2, \quad y = t^3 - 2t^2, \quad t \in \mathbb{R}$$

a Find a Cartesian equation of C in the form $y = f(x)$. **(3 marks)**

b Sketch the curve C. **(3 marks)**

12 Show that the line with equation $y = 4x + 20$ is a tangent to the curve with parametric equations $x = t - 3$, $y = 4 - t^2$. **(4 marks)**

13 The curve C has parametric equations $x = 2\ln t, \quad y = t^2 - 1, \quad t > 0$

a Find the coordinates of the point where the line $x = 5$ intersects the curve. Give your answer as exact values. **(4 marks)**

b Given that the line $y = k$ intersects the curve, find the range of values for k. **(3 marks)**

14 The diagram shows the curve C with parametric equations

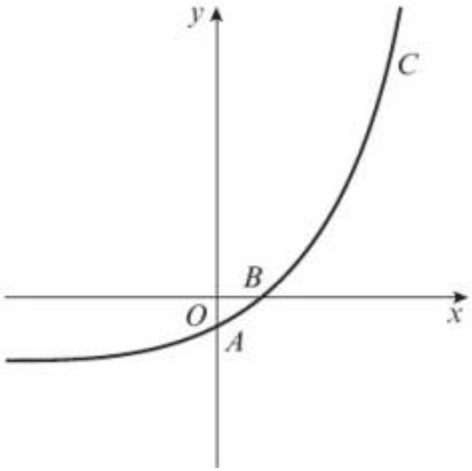
$$x = 1 + 2t, \quad y = 4^t - 1$$

The curve crosses the y-axis and the x-axis at points A and B respectively.

a Find the coordinates of A and B. **(4 marks)**

The line l intersects the curve at points A and B.

b Find the equation of l in the form $ax + by + c = 0$. **(3 marks)**



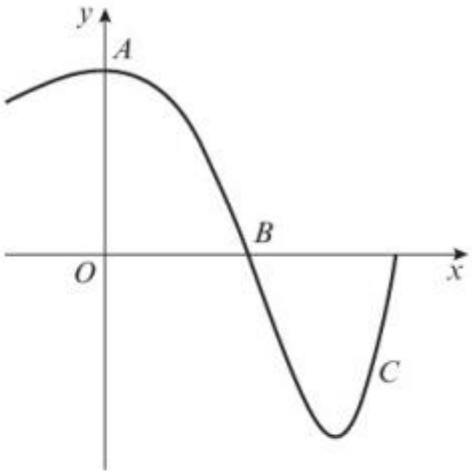
Chapter Exercises

15 The diagram shows the curve C with parametric equations

$$x = \ln t - \ln\left(\frac{\pi}{2}\right), \quad y = \sin t, \quad 0 < t < 2\pi$$

The curve crosses the y-axis and the x-axis at points A and B respectively. The line l intersects the curve at points A and B. Find the equation of l in the form $ax + by + c = 0$.

(7 marks)



16 A plane's position at time t seconds during its descent can be modelled with the following parametric equations

$$x = 80t, \quad y = 3000 - 30t, \quad 0 < t < k$$

where x is the horizontal distance travelled in metres and y is the vertical height of the plane in metres.

a Show that the plane's descent is a straight line. **(3 marks)**

Given that the model is valid until the plane is 30 m off the ground,

b find the value of k **(2 marks)**

c determine the distance travelled by the plane in this portion of its descent. **(3 marks)**

Chapter Exercises

17 The path of an arrow path at time t seconds from being fired can be described using the following parametric equations

$$x = 50\sqrt{2}\,t, \quad y = 1.5 - 4.9t^2 + 50\sqrt{2}\,t, \quad 0 \leqslant t \leqslant k$$

where x is the horizontal distance from the archer in metres and y is the vertical height of the arrow above level ground.

a Find the furthest horizontal distance the arrow can travel.

A castle is located a horizontal distance of 1000 m from the archer's position. The height of the castle is 10 m.

b Show that the arrow misses the castle.

c Find the distance the archer should step back so that he can hit the top of the castle.

18 A mountaineer's climb at time t hours can be modelled with the following parametric equations

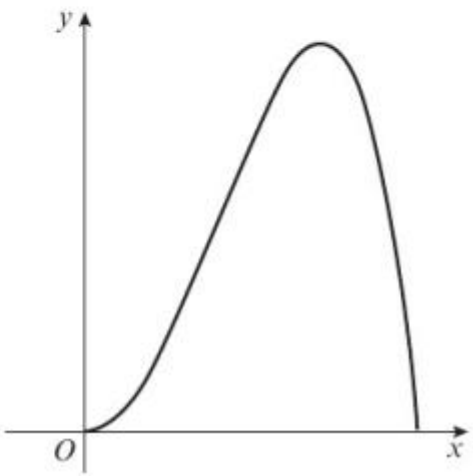
$$x = 300\sqrt{t}, \quad y = 244t(4 - t), \quad 0 < t < k$$

where x represents the distance travelled horizontally in metres and y represents the height above sea level in metres.

a Find the height of the peak and the time at which the mountaineer reaches it. **(3 marks)**

Given that the mountaineer completes her climb when she gets back to sea level,

b find the horizontal distance from the beginning of her climb to the end. **(2 marks)**



Chapter Exercises

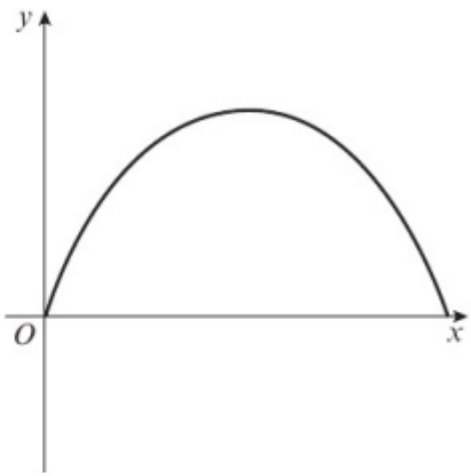
19 A bridge is designed using the following parametric equations:

$$x = \frac{4t}{\pi} - 2\sin t, \quad y = -\cos t, \quad \frac{\pi}{2} < t < \frac{3\pi}{2}$$

Given that 1 unit in the design is 10 m in real life,

a find the highest point of the bridge

b find the width of the widest river this bridge can cross.



20 A BMX cyclist's position on a ramp at time t seconds can be modelled with the parametric equations

$$x = 3(e^t - 1), \quad y = 10(t-1)^2, \quad 0 \leqslant t \leqslant 1.3$$

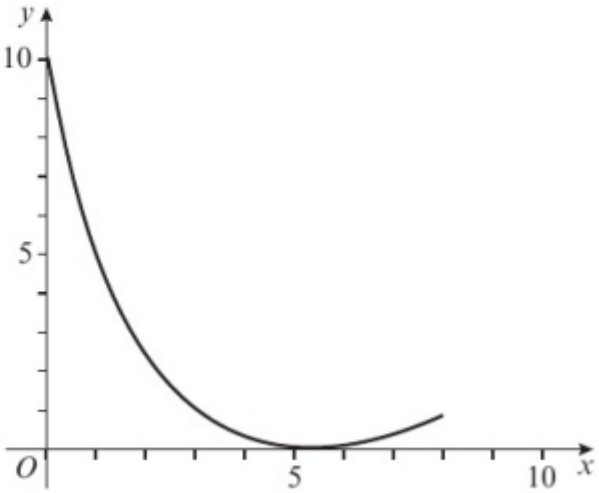
where x is the horizontal distance travelled in metres and y is the height above ground level in metres.

a Find the initial height of the cyclist.

b Find the time the cyclist is at her lowest height.

Given that after 1.3 seconds, the cyclist is at the end of the ramp,

c find the height at which the cyclist leaves the ramp.



Chapter Exercises

Challenge

Two particles A and B move in the x-y plane, such that their positions relative to a fixed origin at a time t seconds are given, respectively, by the parametric equations:

$$A: x = \frac{2}{t}, \quad y = 3t + 1, \quad t > 0$$

$$B: x = 5 - 2t, \quad y = 2t^2 + 2k - 1, \quad t > 0$$

where k is a non-zero constant.

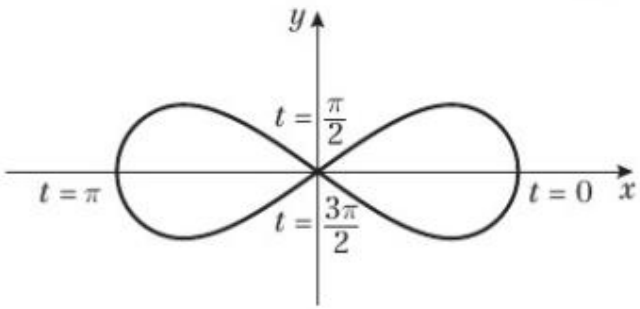
Given that the particles collide,

a find the value of k

b find the coordinates of the point of collision.

1 **a** $A(4, 0), B(0, 3)$ **b** $C\left(2\sqrt{3}, \frac{3}{2}\right)$ **c** $\left(\frac{x}{4}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$

2



3 **a** $y = \ln(2\sqrt{x-1}) - \frac{1}{2}, x > e^3 + 1$

b $y > 1 + \ln 2$

4 $y = -\ln(4) - 2\ln(x), 0 < x < \frac{1}{2}, y > 0$

5 **a** $y = 1 - 2x^2$ **b** $\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$

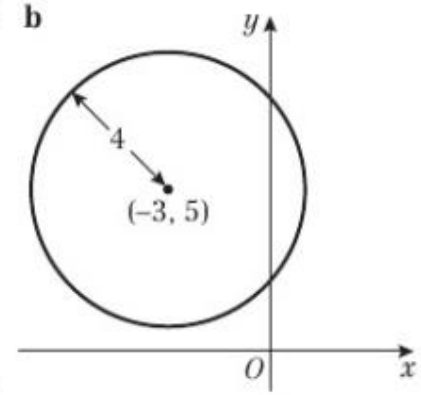
6 $t = \frac{1}{x} - 1$

Sub into $y = \frac{1}{(1+t)(1-t)}$

$$y = \frac{1}{\left(1 + \frac{1}{x} - 1\right)\left(1 - \frac{1}{x} + 1\right)} = \frac{1}{\left(\frac{1}{x}\right)\left(2 - \frac{1}{x}\right)}$$

$$y = \frac{x^2}{x^2\left(\frac{1}{x}\right)\left(2 - \frac{1}{x}\right)} = \frac{x^2}{2x - 1}$$

7 **a** $(x+3)^2 + (y-5)^2 = 16$ **b**



c $(0, 5 + \sqrt{7}), (0, 5 - \sqrt{7})$

8 **a** $x(1+t) = 2 - 3t \Rightarrow xt + 3t = 2 - x \Rightarrow t(x+3) = 2 - x$

$\Rightarrow t = \frac{2-x}{x+3}$

Sub into $y = \frac{3+2t}{1+t}$

$$y = \frac{3 + 2\left(\frac{2-x}{x+3}\right)}{1 + \left(\frac{2-x}{x+3}\right)} = \frac{3(x+3) + 2(2-x)}{x+3+2-x} = \frac{3x+9+4-2x}{5}$$

$$= \frac{x+13}{5} \Rightarrow y = \frac{1}{5}x + \frac{13}{5}$$

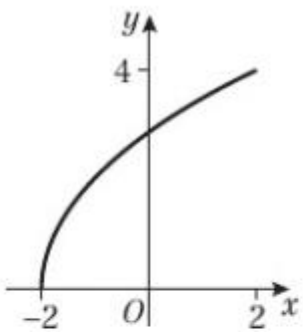
This is in the form $y = mx + c$, therefore the curve C is a straight line.

b $\frac{4\sqrt{26}}{5}$

9 **a** $y = 2\sqrt{x+2}$

b Domain: $-2 \leqslant x \leqslant 2$, Range: $0 \leqslant y \leqslant 4$

c

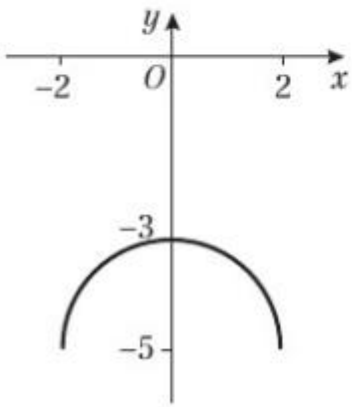


10 **a** $\cos t = \frac{x}{2}$, $\sin t = \frac{y+5}{2}$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y+5}{2}\right)^2 = 1 \Rightarrow x^2 + (y+5)^2 = 4$$

Since $0 \leqslant t \leqslant \pi$, the curve C forms half of a circle.

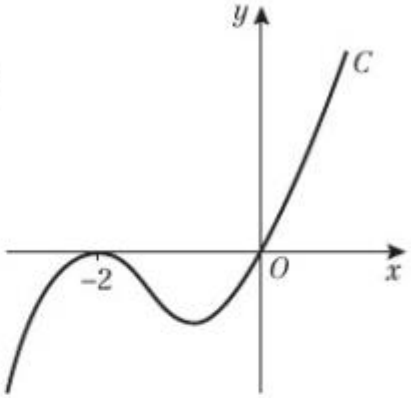
b



c 2π

11 **a** $y = x^3 + 4x^2 + 4x$

b



12 $4 - t^2 = 4(t-3) + 20 \Rightarrow 0 = t^2 + 4t + 4$

Discriminant $= 4^2 - 4 \times 1 \times 4 = 16 - 16 = 0$

So, the line and the curve only intersect once.

Therefore, $y = 4x + 20$ is a tangent to the curve.

13 **a** $(5, e^5 - 1)$ **b** $k > -1$

14 **a** $A\left(0, -\frac{1}{2}\right)$, $B(1, 0)$ **b** $x - 2y - 1 = 0$

15 $x + y \ln 2 - \ln 2 = 0$

16 **a** $t = \frac{x}{80}$, sub into $y = 3000 - 30t$

$y = 3000 - 30\left(\frac{x}{80}\right) \Rightarrow y = 3000 - \frac{3}{8}x$

This is in the form $y = mx + c$, therefore the plane's descent is a straight line.

b $k = 99$ **c** 8458.56 m

17 **a** 1022 m

b $1000 = 50\sqrt{2}\,t \Rightarrow t = 10\sqrt{2}$

Sub into $y = 1.5 - 4.9t^2 + 50\sqrt{2}\,t$

$y = 1.5 - 4.9(10\sqrt{2})^2 + 50\sqrt{2}(10\sqrt{2})$

$y = 21.5$ m

$21.5 > 10$, therefore, the arrow will be too high

c 11.8 m (3 s.f.)

18 **a** 976 m, 2 hours **b** 600 m

19 **a** 10 m **b** 80 m

20 **a** 10 m **b** 1 second **c** 0.9 m

Challenge

a $k = \frac{3}{2}$ **b** $\left(4, \frac{5}{2}\right)$